Arista Access Network Failure Diagnosis and Resolution SOP for KR-PURAM-ACC-24 Node

Node: KR-ACC-24 | **Vendor:** Arista | **Severity:** Minor | **Alarm Code:** ALARM_1882

Description

The failure domain of Optical Transport Failure in the Arista Access Network equipment, specifically on the KR-PURAM-ACC-24 node, is a critical issue that requires immediate attention. This SOP outlines the diagnostic and resolution steps to identify and rectify the root cause of the issue. The Optical Transport Failure can occur due to degradation of SFP/XFP, fiber cuts, or Rx/Tx power loss. The Arista EOS (Extensible Operating System) relies on the OSPF (Open Shortest Path First) protocol for routing and forwarding decisions. In the event of an Optical Transport Failure, the OSPF protocol may not function correctly, leading to network instability and data packet loss. The SFP/XFP degradation or fiber cuts can be caused by aging hardware, environmental factors such as temperature, or physical damage. The Rx/Tx power loss can be caused by faulty SFP/XFP modules or aging fiber optic cables. It is essential to identify the root cause of the issue to prevent further network instability and data packet loss.

Pre-requisite Checks

- Prerequisite: Verify that the KR-PURAM-ACC-24 node is accessible via the management interface, Status: Completed

- Prerequisite: Check the redundant link status to ensure that it is operational, Status: Completed

Potential Root Causes

Cause	Probability
Degradation of SFP/XFP module	High
Fiber cuts or breaks	Medium
Rx/Tx power loss due to faulty SFP/XFP module or aging fiber optic cable	Low

Diagnostic Commands & Expected Outputs

```
Command: show interface status, Expected Output: Interface Status
Protocol Description
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MgmtEth0/RSP0/CPU0/0 Up up management MgmtEth0/RSP0/CPU0/1 Up up
management Eth1/1 Up up unknown , Description: This command displays
the interface status and protocol, which can help identify if the
issue is related to the interface or the protocol.

Command: show sfp detail, Expected Output: SFP Information for
MgmtEth0/RSP0/CPU0/0: ----------------------------------------------
Status: Not Present Type: Not Present Part Number: Not Present Serial
Number: Not Present Optical Power: Not Present SFP Information for
```

```
MgmtEth0/RSP0/CPU0/1: ----------------------------------------------
Status: Present Type: SFP-10G-SR Part Number: HP 1550 nm SMF SFP
Module Serial Number: ABC12345678 Optical Power: 3.5 dBm, Description:
This command displays the SFP/XFP details, which can help identify if
the issue is related to the SFP/XFP module.
```

Resolution Steps

Step 1: Step: Replace the SFP/XFP module, Command: replace sfp interface MgmtEth0/RSP0/CPU0/0, Description: Replace the SFP/XFP module with a new one to resolve the issue.

Step 2: Step: Check the fiber optic cable, Command: show interface MgmtEth0/RSP0/CPU0/1, Description: Check the fiber optic cable for any damage or cuts and replace it if necessary.

Post-Check Verification

```
Command: show interface status, Expected Output: Interface Status
Protocol Description
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MgmtEth0/RSP0/CPU0/0 Up up management MgmtEth0/RSP0/CPU0/1 Up up
management Eth1/1 Up up unknown , Description: Verify that the
interface status and protocol are up and functioning correctly.

Command: show sfp detail, Expected Output: SFP Information for
MgmtEth0/RSP0/CPU0/0: ----------------------------------------------
Status: Present Type: SFP-10G-SR Part Number: HP 1550 nm SMF SFP
Module Serial Number: ABC12345678 Optical Power: 3.5 dBm, Description:
Verify that the SFP/XFP module is functioning correctly and the
optical power is within the specified range.
```

Roll-back Procedures

- Step: Restore the original SFP/XFP module, Command: replace sfp interface MgmtEth0/RSP0/CPU0/0 original, Description: Restore the original SFP/XFP module to revert the changes made during the resolution steps.